Quantum Space-Time Bend of Special Relativity in Riemann Multidimensional Space Derivation Companion — TauTimp (Temporal Undulatory Theory) Project **Author:** Viorel-Mihai Popescu (TauTimp Project) **Source paper:** *Quantum Space-Time Bend of Special Relativity in Riemann Multidimensional Space*, 14May 2026 **Computational support:** Maple 2025 worksheet QmSp-TBend.mpl (Physics package) **This document:** English derivation companion — analytic steps, LaTeX equations, normalization constraints, and the link to the canonical TauTimp wave equation.

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1. Scope and Notation

This companion records the analytic derivations underlying the paper *Quantum Space-Time Bend of Special Relativity in Riemann Multidimensional Space*. The physical thesis is that spacetime curvature is not purely a geometric property of a pseudo-Riemannian manifold but emerges at the quantum level from mediating microparticles, one per physical coordinate of a generalized Riemannian multidimensional space.

Symbol conventions used throughout:

Symbol	Meaning
$f(r,t)$	Effective Onduscular Wave Function (EOWF) of space
R	TauTimp characteristic cosmological radius, $R = 4.2355 \times 10^{22}$ m
R_S	Muon-neutrino cosmological scale parameter (Schwarzschild-analogue for the Local Universe)
r	Radial coordinate
c	Speed of light in vacuum, $c = 299\,792\,458$ m/s
G	Newtonian gravitational constant, $G = 6.6743 \times 10^{-11}$ m^3 kg^{-1} s^{-2}
Δ	Laplacian operator
$\nu,\ \mu,\ \tau$	Electronic, muon, and tau neutrinos (mediators of gravity, space, and time)
P_{To}	Torsion particle (cylindrical-symmetry mediator)
P_γ	Gamma-ray particle
$m = M_\odot$	Solar mass, 1.9885×10^{30} kg

The metric signature is $(-,+,+,+)$, following the Misner–Thorne–Wheeler (MTW) convention. Coordinates are ordered $(x^1, x^2, x^3, x^4) = (ct, r, \theta, \phi)$.

2. The Onduscular Space Wave Equation 2.1 Muon-Neutrino Space Equation — Eq. (1)

The fundamental equation governing quantum space, mediated by the muon neutrino μ, is the variable-coefficient spherical wave equation

$$\frac{R_S}{r}\,\Delta f(r,t) = \frac{1}{c^2}\,\frac{\partial^2 f(r,t)}{\partial t^2}.$$

The coefficient R_S/r replaces the constant unity of the ordinary d'Alembert wave equation, introducing a position-dependent effective medium. In spherical symmetry the Laplacian is

$$\Delta f = \frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\,\frac{\partial f}{\partial r}.$$

2.2 Gravitational (Electronic-Neutrino) Equation — Eq. (3)

The corresponding equation for the gravitational field, mediated by the electronic neutrino ν (the graviton candidate), is structurally identical but uses the cosmological radius R:

$$\frac{R}{r}\,\Delta f(r,t) = \frac{1}{c^2}\,\frac{\partial^2 f(r,t)}{\partial t^2}.$$

2.3 Scaling Relationship Between (1) and (3)

The two equations are connected by the ratio

$$\frac{R_S}{R} = 1979.8.$$

Performing the substitution $r \to r/1979.8$ in Eq. (3) maps it onto Eq. (1):

$$\frac{R}{r/1979.8}\,\Delta f = 1979.8\,\frac{R}{r}\,\Delta f = \frac{R_S}{r}\,\Delta f.$$

Hence the EOWF of space is *proportional but not equal* to the EOWF of the gravitational field. The curvature of space is approximately 1980 times smaller than the curvature produced by the gravitational field — the structural distinction between the μ (space) and ν (gravity) sectors.

3. Separation of Variables and the Airy/Bessel Reduction 3.1 Separation

Insert $f(r,t) = g(r)\,u(t)$ into Eq. (3). Dividing by $g(r)\,u(t)$,

$$\frac{R}{r}\frac{1}{g}\left(g'' + \frac{2}{r}g'\right) = \frac{1}{c^2}\frac{\ddot{u}}{u} = -k^2,$$

with $k^2 > 0$ the separation constant. The temporal equation is the harmonic oscillator

$$\ddot{u}(t) + c^2k^2\,u(t) = 0 \quad \Rightarrow \quad u(t) = A\cos(ckt) + B\sin(ckt).$$

The radial equation is

$$g''(r) + \frac{2}{r}\,g'(r) + \frac{k^2}{R}\,r\,g(r) = 0.$$

3.2 Reduction via $g = \psi/r$

Substituting $g(r) = \psi(r)/r$ removes the first-derivative term:

$$g' = \frac{\psi'}{r} - \frac{\psi}{r^2}, \qquad g'' = \frac{\psi''}{r} - \frac{2\psi'}{r^2} + \frac{2\psi}{r^3},$$

so that $g'' + 2/r\,g' = \psi''/r$. Equation (3a) becomes the **Airy-type equation**

$$\psi''(r) + \frac{k^2}{R}\,r\,\psi(r) = 0.$$

3.3 Airy Form

Introduce the affine rescaling

$$\xi = -\left(\frac{k^2}{R}\right)^{1/3} r,$$

under which Eq. (3b) takes the canonical Airy form

$$\frac{d^2\psi}{d\xi^2} - \xi\,\psi = 0 \quad \Rightarrow \quad \psi(\xi) = c_1\,\mathrm{Ai}(\xi) + c_2\,\mathrm{Bi}(\xi).$$

Regularity of $g = \psi/r$ at the origin discards the Bi branch (and the part of Ai that does not vanish appropriately), leaving

$$g(r) = \frac{c_1}{r}\,\mathrm{Ai}\left[-(k^2/R)^{1/3}r\right].$$

3.4 Equivalent Bessel-1/3 Representation

Equation (3a) may instead be solved directly. With $g(r) = r^{-1/2}\,w(r)$ and $s = 2/3\,\sqrt{k^2/R}\;r^{3/2}$, the function w satisfies Bessel's equation of order $1/3$: $g(r) = r^{-1/2}\left[\,c_1\,J_{1/3}(s) + c_2\,J_{-1/3}(s)\,\right], \qquad s = \frac{2}{3}\sqrt{\frac{k^2}{R}}\;r^{3/2}.$

The Airy and Bessel-$1/3$ representations are identical, related by the standard identity $\mathrm{Ai}(-x) = 1/3\sqrt{x}\,[\,J_{1/3}(\zeta) + J_{-1/3}(\zeta)\,]$ with $\zeta = 2/3\,x^{3/2}$.

3.5 Eigenvalue Quantization

The boundary condition $g(2R_S) = 0$ (Section 4) selects a discrete spectrum. Writing a_n for the n-th negative zero of the Airy function, $\mathrm{Ai}(a_n) = 0$, the quantization condition is

$$-\left(\frac{k_n^2}{R}\right)^{1/3}(2R_S) = a_n \quad \Rightarrow \quad k_n^2 = R\,\frac{|a_n|^3}{(2R_S)^3}.$$

The associated temporal frequencies are $\omega_n = c\,k_n$, and the eigenfunctions are

$$f_n(r,t) = \frac{c_1}{r}\,\mathrm{Ai}\left[-(k_n^2/R)^{1/3}r\right]\left[A_n\cos(\omega_n t) + B_n\sin(\omega_n t)\right].$$

4. Boundary Conditions and the Cosmological Scale Parameters 4.1 The Three Determining Conditions

Three independent conditions fix the basis of separated solutions:

Condition	Expression	Physical meaning
A	$f(2R_S,t) = 0$ for all t (equivalently $g(2R_S) = 0$)	The EOWF vanishes at the cosmological boundary $r = 2R_S$
B	$g'(0) = R$	The initial radial derivative at the origin equals the cosmological radius R
C	speed of interaction $= c$	Massless propagation, via the divergence condition Eq. (2)

4.2 Derivation of $2R_S$

The boundary radius is linked to the nuclear scale through the structural ratio of the nucleon and quark dimensions:

$$2R_S = R \cdot \frac{\dim(\text{nucleon})}{\dim(\text{quark})} = 2 \times 4.2355 \times \frac{0.841419}{0.43} \times 10^{25}\ \text{m}.$$

Numerically,

$$\frac{0.841419}{0.43} \approx 1.9568, \qquad 2R_S \approx 16.7710 \times 10^{25}\ \text{m} \approx 17.7274 \text{ billion light-years}.$$

The proton charge radius ≈ 0.8414 fm (CODATA) and the quark confinement radius ≈ 0.43 fm (QCD confinement scale) thus provide a sub-nuclear anchor for a cosmological-scale boundary — the structural bridge between the smallest and largest scales asserted by the framework.

5. The Schwarzschild Metric and Einstein's Field Equations 5.1 Einstein's Field Equation — Eq. (4)

$$R_{\mu\nu} - 1/2\ R_c\, g_{\mu\nu} + \Lambda\, g_{\mu\nu} = -\frac{8\pi G}{c^4}\, T_{\mu\nu} = -K\, T_{\mu\nu},$$

with the coupling constant

$$K = \frac{8\pi G}{c^4} = 2.076647 \times 10^{-43}\ \text{N}^{-1} \quad [\text{s}^2\,\text{m}^{-1}\,\text{kg}^{-1}].$$

Each term carries dimension $[\,L^{-2}\,]$. The sign convention $-K\,T_{\mu\nu}$ on the right-hand side follows MTW with signature $(-,+,+,+)$.

5.2 The Schwarzschild Metric Tensor

With $m = M_\odot$ and coordinates (ct, r, θ, ϕ),

$$g_{\mu\nu} = \begin{pmatrix} -\left(1 - \frac{2Gm}{rc^2}\right) & 0 & 0 & 0 \\ 0 & -r^2 & 0 & 0 \\ 0 & 0 & -r^2\sin^2\theta & 0 \\ 0 & 0 & 0 & \left(1 - \frac{2Gm}{rc^2}\right)^{-1} \end{pmatrix}.$$

The line element — Eq. (5) — is

$$ds^2 = -\frac{rc^2 - 2Gm}{r}\, dt^2 + r^2\, d\theta^2 + r^2\sin^2\theta\, d\Phi^2 + \frac{rc^2}{rc^2 - 2Gm}\, dr^2,$$

with $e^{2\phi} = e^{-2\Lambda} = \left(1 - \frac{2Gm}{rc^2}\right)$, where ϕ and Λ are functions of r alone. Identifying the Schwarzschild radius $R_S = 2Gm/c^2$ recovers the textbook form

$$ds^2 = -\left(1 - \frac{R_S}{r}\right) c^2\, dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2.$$

The vacuum solution satisfies $R_{\mu\nu} = 0$, $R_c = 0$, $\Lambda = 0$.

6. Coordinate Transformations and the Metric Variants 6.1 First Transformation (TransformCoordinates) — Eq. (5.1)

With the notation $R_{\text{Schwar}} = R_S$ and $R_S = 2Gm/c^2$, define the coordinate change

$$\text{TR:} \quad r \;\mapsto\; 1 - \frac{2\rho}{R_S} = \frac{mG - \rho c^2}{mG}.$$

Under TR the metric becomes

$$ds_{21}^2 = -\frac{c^6\,(mG - rc^2)}{m^2G^2\,(c^4r - 2mGc^2 + 2m^2G^2)}\, dr^2 + \frac{(mG - rc^2)^2}{m^2G^2}\, d\theta^2 + \frac{(mG - rc^2)^2}{m^2G^2}\sin^2\theta\, d\Phi^2 + \frac{c^4r - 2mGc^2 + 2m^2G^2}{mG - rc^2}\, dt^2.$$

For $T_{\mu\nu} = 0$ and $\Lambda = 0$ the field equation reduces to the identity $0 = 0$ — a coordinate transformation does not alter the physical content of the vacuum Schwarzschild solution. This is verified symbolically in the Maple worksheet.

6.2 Inverse Transformation — Eq. (6)

Applying the inverse change of variable $\rho = mG\,(1 - r)/c^2$ and re-expressing the metric as a function of r alone yields

$$ds_{22}^2 = -\frac{c^6 r}{m^2G^2\,(2mG - rc^2)}\, dr^2 + r^2\, d\theta^2 + r^2\sin^2\theta\, d\Phi^2 + \frac{2mG - rc^2}{r}\, dt^2.$$

For this metric the Ricci scalar and cosmological constant are **non-zero**:

$$R_{cp} = 2\,\frac{c^4 - G^2m^2}{c^4r^2}, \qquad \Lambda_p = \frac{c^4 - G^2m^2}{12\,c^4r^2}.$$

The modified metric tensor and proportional constant are

$$g_{\mu p} = \begin{pmatrix} \dfrac{rc^6}{G^2m^2(rc^2+2Gm)} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2\sin^2\theta & 0 \\ 0 & 0 & 0 & \dfrac{-rc^2-2Gm}{r} \end{pmatrix}, \qquad K_p = \frac{G}{c^2} = 7.42616 \times 10^{-28}\ \text{[m/kg]}.$$

Metric (6) is *not* the standard Schwarzschild vacuum solution; it arises from a specific non-trivial change of variables. The condition $T_{\mu p} = 0$ requires $Gm = c^2$, i.e. $m = c^2/G \approx 1.35 \times 10^{27}$ kg, about 675 times smaller than $M_\odot$. The physical reading is that the γ-ray and torsion-field curvature is roughly 675 times smaller than the gravitational-field space curvature generated by the Sun.

6.3 Re-scaled Mass Substitution

With the substitution $m = 2M/R_S$ (M a re-scaled mass parameter), the cosmological constant becomes

$$\Lambda_\tau = \frac{c^4R_S^2 - 4G^2M^2}{12\,R_S^2\,c^4\,r^2},$$

and the corresponding metric tensor is

$$g_{\mu p} = \begin{pmatrix} \dfrac{-R_S^3c^6r}{4\,(-rR_Sc^2+4GM)\,G^2M^2} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2\sin^2\theta & 0 \\ 0 & 0 & 0 & \dfrac{-rR_Sc^2+4GM}{rR_S} \end{pmatrix}.$$

The eigenvalues of the characteristic polynomial yield the closure condition $c^2 = mG$, which is exactly what the substitution $m = 2M/R_S$ enforces, producing the null tensor $T_{\mu p} = 0$.

7. The Torsion Particle and the Second Field-Equation Variant

The torsion particle P_{To} — associated with the synchronized precession of an ensemble of spin-bearing particles — introduces a second set of field equations with **cylindrical symmetry**:

$$R_{\mu p} - 1/2\ R_{cp}\,g_{\mu p} + \Lambda\,g_{\mu p} = -K_p\,T_{\mu p}, \qquad K_p = \frac{G}{c^2} = 7.42616 \times 10^{-28}\ \text{[m/kg]}.$$

This parallels Einstein's Eq. (4) but with a distinct Ricci tensor $R_{\mu p}$, scalar R_{cp}, cosmological constant Λ_p, and coupling K_p (dimensionally [m/kg] rather than [N^{-1}]). The boundary condition is $f(2R, t) = 0$, the same form as the gravitational ν case, but applied in cylindrical rather than spherical geometry.

Side-by-side comparison:

Quantity	Gravity (ν)	Torsion (P_{To})
Coupling	$K = 8\pi G/c^4 = 2.077 \times 10^{-43}\ \text{N}^{-1}$	$K_p = G/c^2 = 7.426 \times 10^{-28}$ m/kg
Symmetry	Spherical	Cylindrical
Ricci scalar	R_c (standard)	$R_{cp} = 2(c^4 - G^2m^2)/(c^4r^2)$
Cosmological constant	$\Lambda = 0$ (vacuum)	$\Lambda_p \neq 0$

Because each coordinate bends spacetime through a **first-order linear** equation, the total spacetime curvature is treated as **cumulative (superposable)** — the contributions of the ν, τ, μ, P_{To} and P_γ sectors add. This superposition holds in the linearized (weak-field) regime, the same regime in which gravitational-wave theory is formulated.

8. Attraction/Repulsion Transition and Observational Anomalies 8.1 Transition Scale

The γ-ray particle P_γ and the torsion particle P_{To} are proposed as the agents of cosmic expansion, with an attraction/repulsion transition at

$$2R \approx 8.954 \text{ million light-years} \approx 8.471 \times 10^{22}\ \text{m}.$$

- For $r < 2R$: P_{To} generates **attraction** between galaxies.
- For $2R < r < 2R_S \approx 17.727$ billion light-years: P_{To} generates **repulsion**, playing the role of dark energy.

The transition scale is set directly by the canonical TauTimp radius $R = 4.2355 \times 10^{22}$ m, since $2R = 8.471 \times 10^{22}$ m.

8.2 Cited Observational Anomalies

The cumulative ($\nu + P_{To}$) bending is offered as the origin of three astronomical anomalies:

1. **Venus trajectory anomaly** — a cumulative-curvature effect known in astronomical circles.
2. **Halley's Comet perihelion delay (1910)** — a 3-day delay at perihelion, cited from Prof. Dr. Ioan N. Popescu, *Gravity* (1982), p. 184.
3. **Mercury perihelion advance** — observed at $42.6'' \pm 0.9''$ per century (Popescu, *Gravity*, p. 378), against the standard general-relativistic value of $\approx 43''$ per century.

9. Consistency with the Canonical TauTimp Framework

The canonical TauTimp wave equation is the variable-coefficient spherical PDE

$$\frac{R}{r}\,\Delta f = \frac{1}{c^2}\frac{\partial^2 f}{\partial t^2}, \qquad R = 4.2355 \times 10^{22}\ \text{m},$$

which is exactly Eq. (3) of the present paper. The space-bend formulation thus inherits the canonical TauTimp infrastructure without modification:

- **Same operator structure** — the $(R/r)\Delta$ coefficient, separation of variables, and the Airy reduction $g = \psi/r$ are identical to the established TauTimp derivation pipeline.
- **Same eigenstates** — the Airy-function eigenstates with quantization from the zeros a_n of Ai carry over directly.
- **Same boundary radius** — $R = 4.2355 \times 10^{22}$ m (≈ 4.5 Mly) is the canonical TauTimp cosmological boundary parameter; $2R$ is the attraction/repulsion transition.
- **New parameter** — the muon-sector scale R_S, related by $R_S/R = 1979.8$, is specific to the space-bend paper. Equation (1) is the μ-sector counterpart of the canonical ν-sector Eq. (3), and the substitution $r \to r/1979.8$ maps one onto the other.

In short, Eq. (3) *is* the canonical TauTimp equation; Eq. (1) is its rescaled muon-sector partner; and the Schwarzschild/torsion analysis of Sections 5–7 supplies the general-relativistic embedding for the same framework.

10. Summary of Key Results

Result	Expression	Value
Characteristic radius	R	4.2355×10^{22} m
Muon-neutrino cosmological scale	R_S	$R_S = R \cdot 1979.8/2 \approx 4.193 \times 10^{25}$ m
Observable-universe boundary	$2R_S$	≈ 17.727 Gly
Spatial/gravitational curvature ratio	R_S/R	≈ 1979.8
Schwarzschild coupling	$K = 8\pi G/c^4$	$2.0766 \times 10^{-43}\ \text{N}^{-1}$
Torsion coupling	$K_p = G/c^2$	7.4262×10^{-28} m/kg
Attraction/repulsion transition	$2R$	$\approx 8.471 \times 10^{22}$ m ≈ 8.95 Mly
Ricci scalar (metric 6)	R_{cp}	$2(c^4 - G^2m^2)/(c^4r^2)$
Cosmological constant (metric 6)	Λ_p	$(c^4 - G^2m^2)/(12\,c^4r^2)$
Re-scaled cosmological constant	Λ_τ	$(c^4R_S^2 - 4G^2M^2)/(12\,R_S^2c^4r^2)$
Radial eigenvalue spectrum	k_n^2	$R\,\lvert a_n\rvert^3/(2R_S)^3,\ \text{Ai}(a_n) = 0$

11. References

[1] MIT, *Einstein's Field Equation — Schwarzschild Solution Derivation*, pp. 7–10. http://web.mit.edu/klmitch/classes/8.033/Schwarzschild.pdf

[2] C. W. Misner, K. S. Thorne, J. A. Wheeler, *Gravitation*, W. H. Freeman, 1973 (copyright 1970–1971).

[3] Prof. Dr. Ioan N. Popescu, *Gravity (Gravitația)*, Editura Științifică și Tehnică, Bucharest, 1982.

Computational links (author's site): - Schwarzschild AXR result: https://michaelvio.github.io/SchwarAXR.pdf - Sun metric: https://michaelvio.github.io/Sun.pdf - Proportional constant derivation: https://michaelvio.github.io/ConstProp.pdf - Tau bending: https://michaelvio.github.io/TauBend.pdf - Gravitational flux tau: https://michaelvio.github.io/GFtau.pdf - Gravity bending overview: https://michaelvio.github.io/GravBend.pdf - viXra preprint: https://vixra.org/pdf/1511.0101v1.pdf https://michaelvio.github.io/QTBendP.pdf

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